

NILOY JANA

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ABOUT ME

AI/ML Engineer with hands-on experience designing and orchestrating multi-agent LLM systems and agentic AI workflows, including production agent pipelines and RAG-based reasoning systems. Skilled in Python, LangChain, and FastAPI, with a track record of building autonomous agents that automate real enterprise workflows and deliver measurable accuracy and efficiency gains.

EDUCATION

St. Thomas College of Engineering and Technology - B.Tech, Computer Science and Engineering (AI/ML) **2027**
South Point High School - CBSE X | CBSE XII (Physics, Chemistry, Mathematics, Biology) **2021**

EXPERIENCE

AI/ML Engineer Intern, KreupAI Technologies LLC **Apr 2026 - Jul 2026**

- Developed a multi-agent LLM system using Python and FastAPI, orchestrating 25 AI agents, including an AI AP Officer and AI QA Coordinator, to automate enterprise QA and finance workflows, monitoring performance across ~2M tokens and reducing API costs by ~18%.
- Built an AI-powered invoice processing agent using NLP and document parsing to perform automated 3-way matching and fraud detection, reducing manual processing by ~75%.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using text embeddings and pgvector in PostgreSQL for semantic search and context-aware AI responses.

Artificial Intelligence Intern, Pratinik Infotech **Mar 2026 - May 2026**

- Built a predictive maintenance ML pipeline in Python, engineering time-series features (rolling stats, lag variables, degradation trends) from sensor data and applying anomaly detection to achieve ~89% failure-prediction accuracy, reducing equipment downtime by ~30%.
- Developed a real-time monitoring pipeline with an interactive health-metrics dashboard, identifying key failure-driving features through feature importance analysis and enabling proactive maintenance recommendations across equipment failure and anomaly detection use cases.

PROJECTS

ARIA - Multi-Agent AI Financial Advisor | Python, FastAPI, RAG (FAISS), SQLite

- Designed a sequential multi-agent LLM pipeline (Auditor, Market Intelligence, Manager agents) synthesizing internal and live market data into one advisory report, backed by a 5-minute cache layer cutting redundant external API calls.
- Implemented a RAG pipeline (LangChain, FAISS, HuggingFace embeddings) with 800-token chunking and top-4 semantic retrieval, paired with a dual local (Ollama) / cloud (Groq) inference architecture for privacy-first, offline-capable AI.
- Built the supporting FastAPI backend exposing 31 REST endpoints across 8 financial modules (retirement planning, tax optimization, portfolio analysis, risk profiling), backed by SQLite persistence.

Opsira - AI Eye Disease Detection System | Python, TensorFlow/Keras, OpenCV, Flask

- Developed a computer vision and deep learning eye health screening system using CNN (EfficientNetB0) with 2-phase transfer learning, classifying 4 ocular conditions from 224x224 medical images with data augmentation and balanced class weighting.
- Implemented Explainable AI using Grad-CAM heatmaps to visualize model decision regions, and built a rule-based OpenCV pipeline extracting 5 physiological biomarkers with a hybrid boosting layer and 3-tier severity classification.

TECHNICAL SKILLS

AI/ML: Agentic AI, Generative AI, RAG, Deep Learning, NLP (Transformers, BERT, LLMs), Multi-Agent Systems

Languages: Python, Java

Frameworks & Libraries: LangChain, TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, OpenCV

Web Development: HTML, CSS, JavaScript, React, FastAPI, Flask, Node.js, REST APIs

Databases: MySQL, PostgreSQL, SQLite

Tools: Git, GitHub

CERTIFICATIONS

- Infosys Certified in Deep Learning** - Demonstrating proficiency in AI/ML concepts and practical model development **Apr 2026**
- IBM AI Engineering Professional Certificate** - Comprehensive training in Machine Learning, Deep Learning, Generative AI, LLMs, RAG, Transformers, and AI Agent Development **May 2026**

ACHIEVEMENTS

- AI Hackathon, IIT Delhi hosted by Eightfold AI: Ranked Top 8 of 1,400+ participants; built a hybrid LLM and rule-based AI agent for a simulation environment.
- Led organization and execution of a college-level coding competition, managing participants and event coordination.
- Reached the semifinals among 1,100 participants in the ET-AI Hackathon hosted by The Economic Times.